Motivation

My introduction

I am Dhruv Savla, a first year UG student in the mechanical department. I like to learn and make functional things. I have made an XLR8 bot with my team and in my current makerspace project I am working on a joystick-controlled mechanical arm.

Why hackerfab?

Hackerfab's goal is to make very tiny (~10um) scale manufacturing accessible to students. I am interested in making MEMS devices, it has huge scope in many fields- medicine, fabrication (DLPs), sensors and control. I am also interested in learning lithography and etching processes because they are crucial parts in my family's pcb making business (Chain Electronics Private Limited). We also have a venture in making uv exposure machines (https://phinixlighting.com/).

Short-Term Goals

Take the 4" precision hot plate from my design to physical realization in the lab for repeatable photoresist soft-baking and hard-baking/annealing.
Make ESC chuck
Learn the use of workshop tools. (not being stuck with cardboard and fevicol and 3d printing for making anything is delightful)

Long-Term Goals

Learn microfabrication and make something useful out of it. Work towards the 'Fab-in-a-box' goal by streamlining the designs and finding the right component suppliers so that other colleges can also start their hackerfabs at a lower cost.